ELECTRICITY CONSUMPTION TIME SERIES FORECASTING USING LINEAR, NON-LINEAR, DEEP LEARNING & HYBRID MODELS

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Geshna B - CB.SC.U4AIE23360
Malavika S Prasad - CB.SC.U4AIE23315
Vibhu Sanchana - CB.SC.U4AIE23347
Asmi K - CB.SC.U4AIE23351

INTRODUCTION

- Electricity consumption forecasting is important for effective power management, cost control, and reliable electricity supply. Accurate prediction of electricity usage helps power companies and consumers plan energy usage efficiently and avoid wastage. However, electricity consumption data changes over time and is influenced by daily habits, seasons, and sudden variations, making forecasting a challenging task.

- This project focuses on electricity consumption time series forecasting using different approaches such as linear models, non-linear models, deep learning models, and hybrid methods. The electricity usage data is first cleaned and analyzed to understand patterns like trends and seasonality. Basic statistical models such as ARIMA are used to predict future consumption, while GARCH is applied to analyze changes in consumption variability.

- The project also explores advanced methods like LSTM and GRU networks, which are capable of learning complex patterns in time-based data. Overall, this project demonstrates how different forecasting techniques can be applied to predict electricity consumption more effectively.

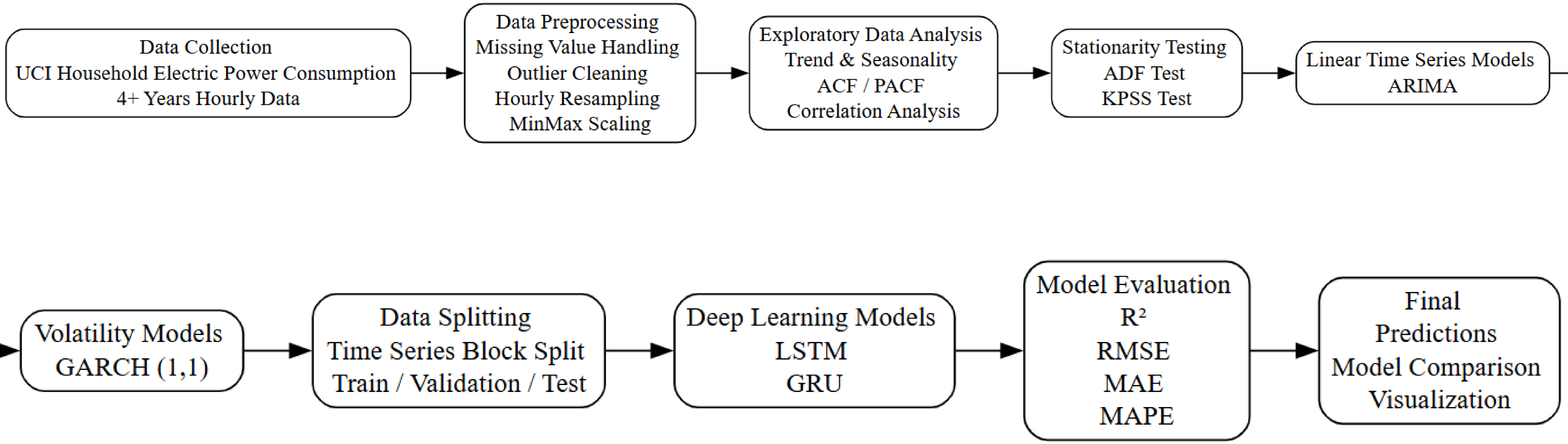
PROBLEM STATEMENT

The given electricity consumption data is a time series, where power usage values are recorded continuously over time. The main problem addressed in this project is forecasting future electricity consumption using historical usage data.

Electricity demand changes due to daily routines, seasonal effects, and user behavior, which makes accurate prediction difficult. By modeling this time series data, the project aims to identify underlying patterns, trends, and variations and use them to predict future consumption values.

The motivation for modeling this time series comes from its real-world importance in power systems. Accurate electricity consumption forecasting helps in efficient energy planning, load management, reducing power wastage, and ensuring a stable electricity supply. The expected outcome of this work is to build reliable forecasting models that can assist energy providers and consumers in making better decisions.

WORKFLOW


Data Collection
UCI Household Electric Power Consumption
4+ Years Hourly Data
Data Preprocessing
Missing Value Handling
Outlier Cleaning
Hourly Resampling
MinMax Scaling
Exploratory Data Analysis
Trend & Seasonality
ACF / PACF
Correlation Analysis
Stationarity Testing
ADF Test
KPSS Test
Linear Time Series Models
ARIMA
Volatility Models
GARCH (1,1)
Data Splitting
Time Series Block Split
Train / Validation / Test
Deep Learning Models
LSTM
GRU
Model Evaluation
R²
RMSE
MAE
MAPE
Final
Predictions
Model Comparison
Visualization

DATASET

NAME: HOUSEHOLD ELECTRICITY POWER CONSUMPTION DATASET
SOURCE: UCI MACHINE LEARNING REPOSITORY
DURATION: 2006 – 2010 (NEARLY 4 YEARS)
TIME FREQUENCY: MINUTE-LEVEL DATA (LATER CONVERTED TO HOURLY AND DAILY)
TYPE: REAL-WORLD HOUSEHOLD ELECTRICITY USAGE DATA

WHY THIS DATA IS TIME SERIES:

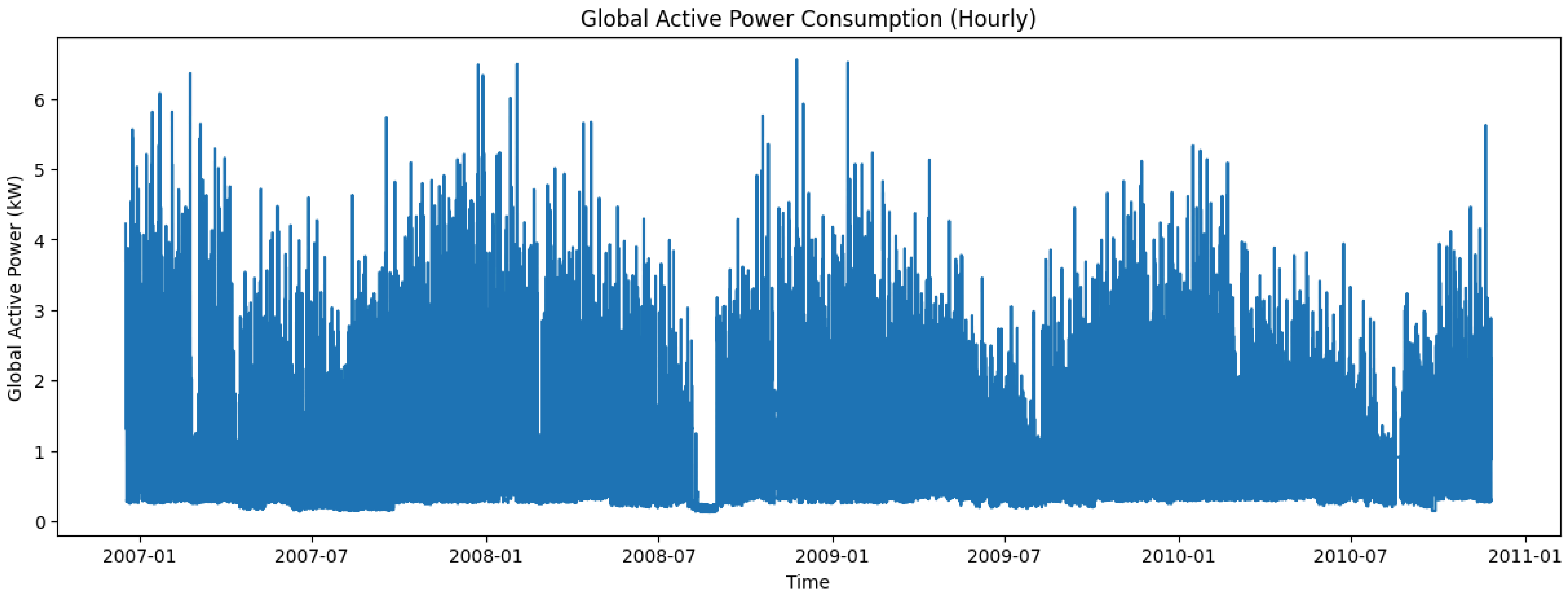
- DATA IS RECORDED CONTINUOUSLY OVER TIME
- EACH VALUE IS LINKED TO A SPECIFIC TIMESTAMP (DATE & TIME)
- CURRENT ELECTRICITY USAGE DEPENDS ON PAST USAGE PATTERNS
- SHOWS TREND, SEASONALITY, AND FLUCTUATIONS, WHICH ARE CORE PROPERTIES OF TIME SERIES DATA

EXPLORATORY DATA ANALYSIS (EDA)

EDA is the initial investigation of our data to discover patterns, spot anomalies, and check assumptions with the help of summary statistics and graphical representations. It’s essentially "getting to know your data" before you try to model it.

Why did we use it?

- Sanity Check: To ensure the data was loaded correctly (no weird dates, no impossible values).
- Pattern Recognition: To visually see if there is a Trend or Seasonality.
- Outlier Detection: To spot spikes or dips that shouldn't be there (e.g., data entry errors or grid failures).
- Distribution Check: To see if the data follows a normal distribution (bell curve), which helps in choosing the right statistical tests later.

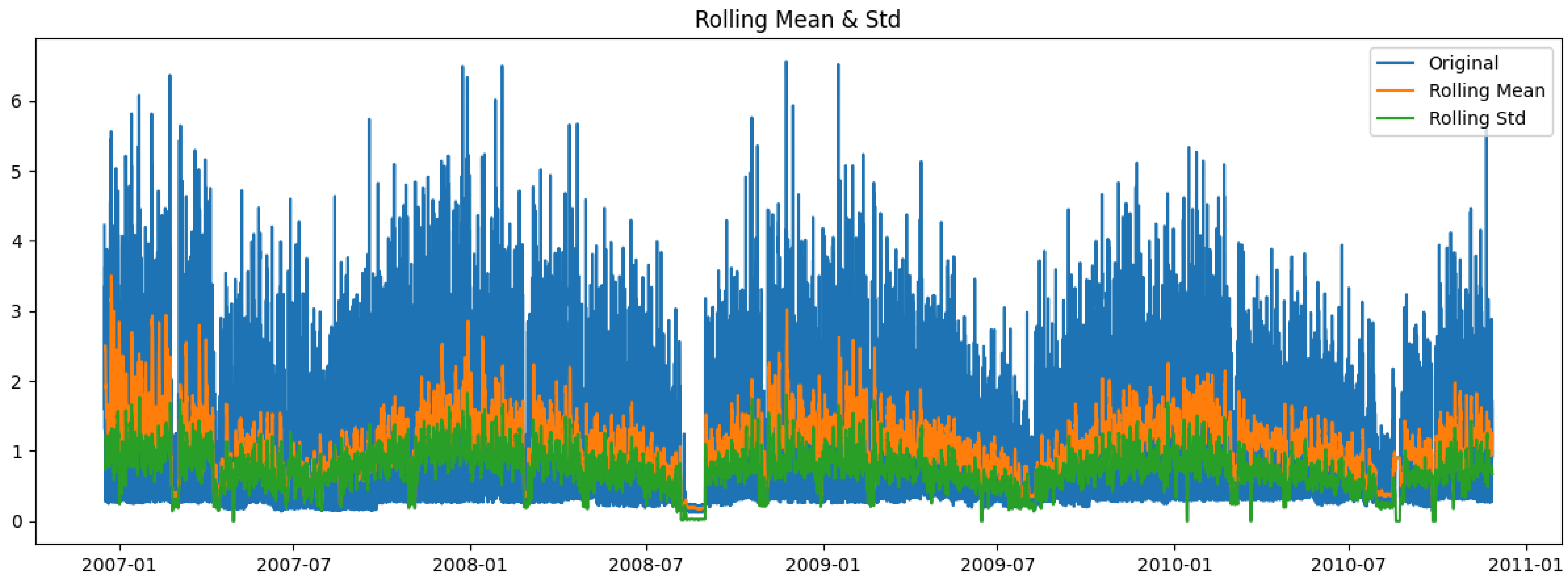


visually inspect the data for:

Trend: Is the power consumption increasing or decreasing over time?

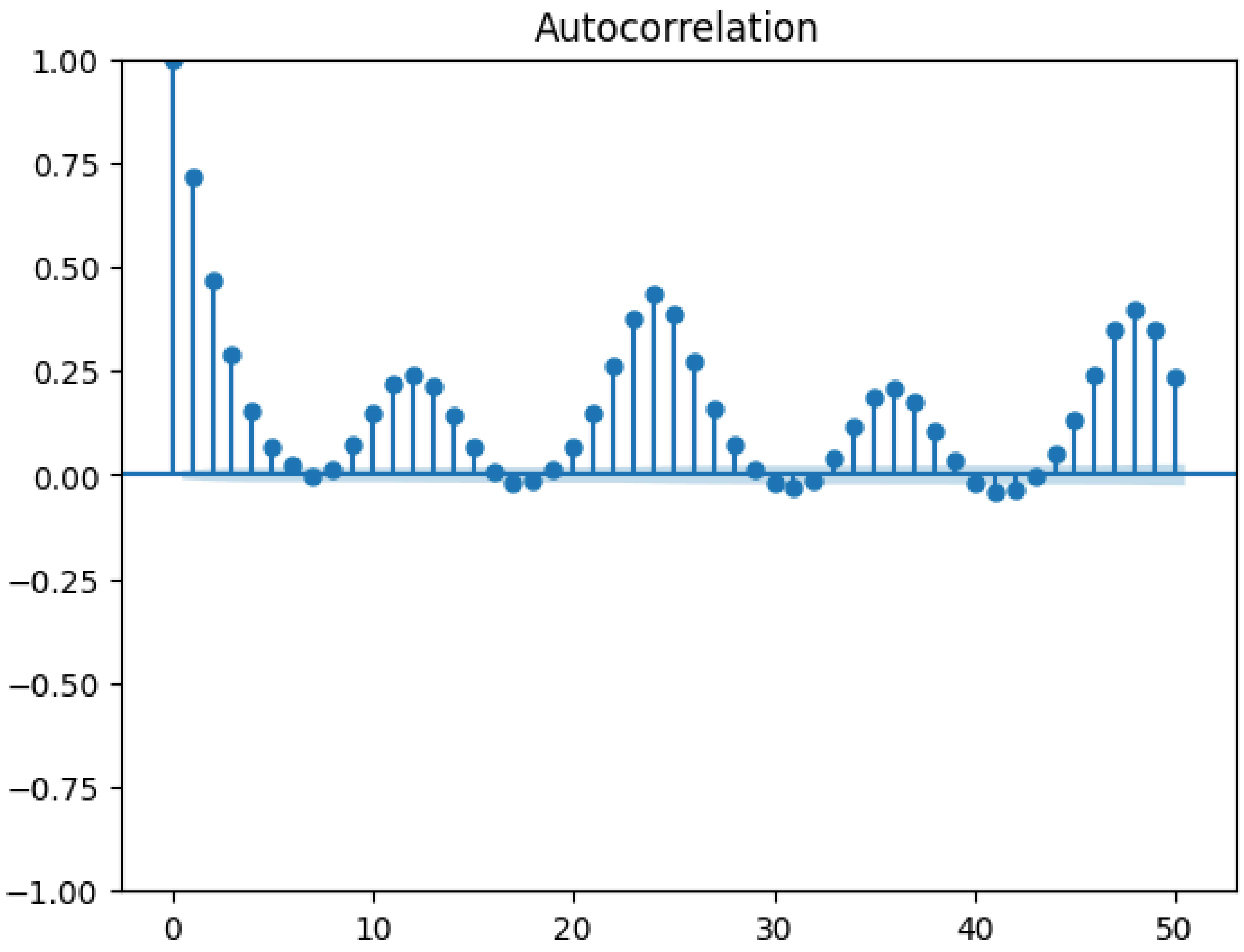
Seasonality: Are there repeating patterns (e.g., weekly or daily peaks)?

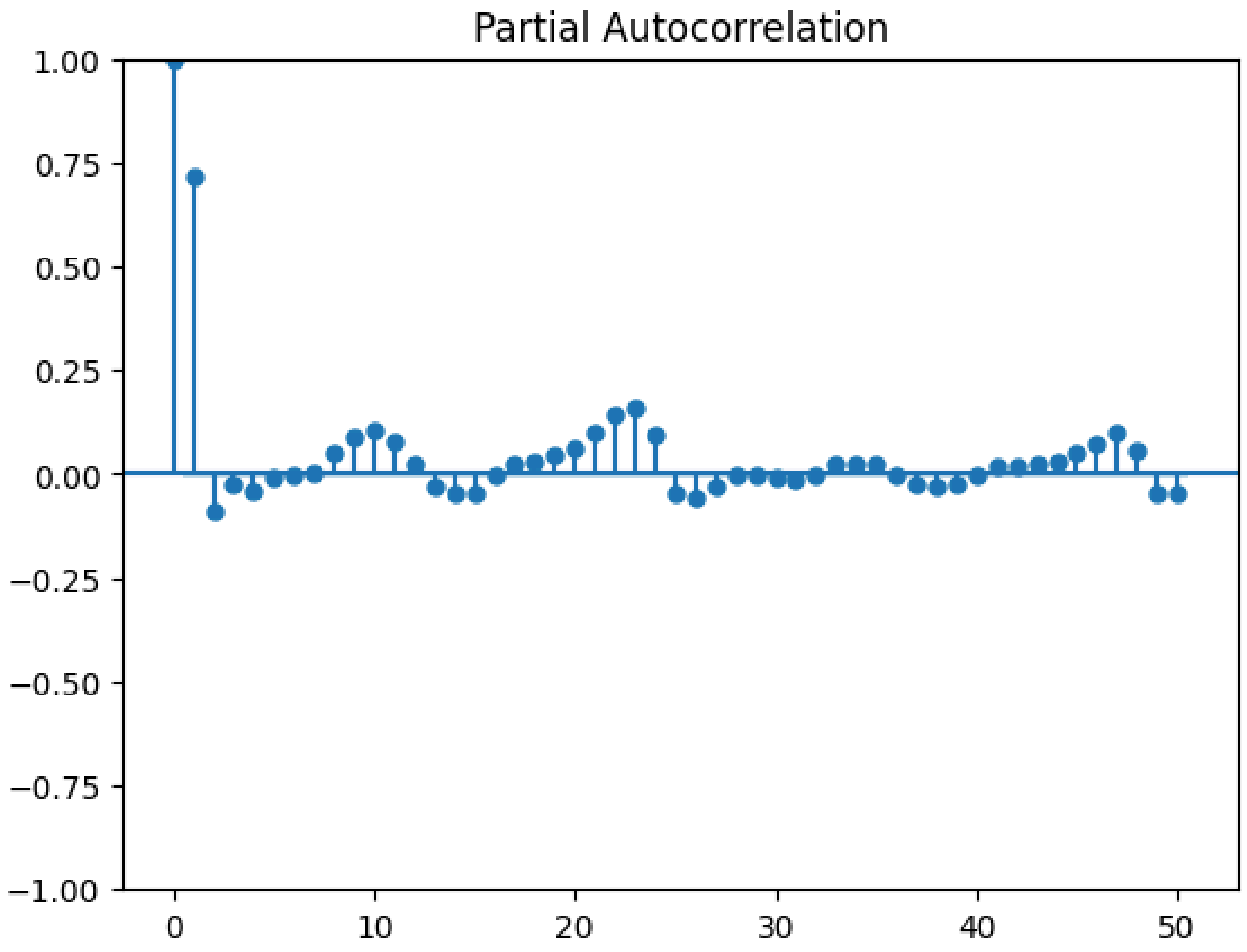
Structural Breaks/Outliers: Are there sudden spikes or drops that indicate anomalies (like the large gaps or spikes seen in the data)?



visual test for Stationarity.

- Stationarity Check: For a time series to be stationary (ideal for modeling), the mean and standard deviation should remain roughly constant over time (straight lines).
- If you see the Rolling Mean trending up/down or the Rolling Std expanding, the data is non-stationary and requires transformation (like differencing) before modeling.





critical for determining the parameters of forecasting models like ARIMA:

ACF (AutoCorrelation Function): Shows how correlated the current value is with past values (lags). It helps identify Seasonality (repeating spikes at specific intervals) and the Moving Average (q) parameter.

PACF (Partial ACF): Shows the direct correlation with past lags after removing intermediate effects. It helps identify the AutoRegressive (p) parameter.

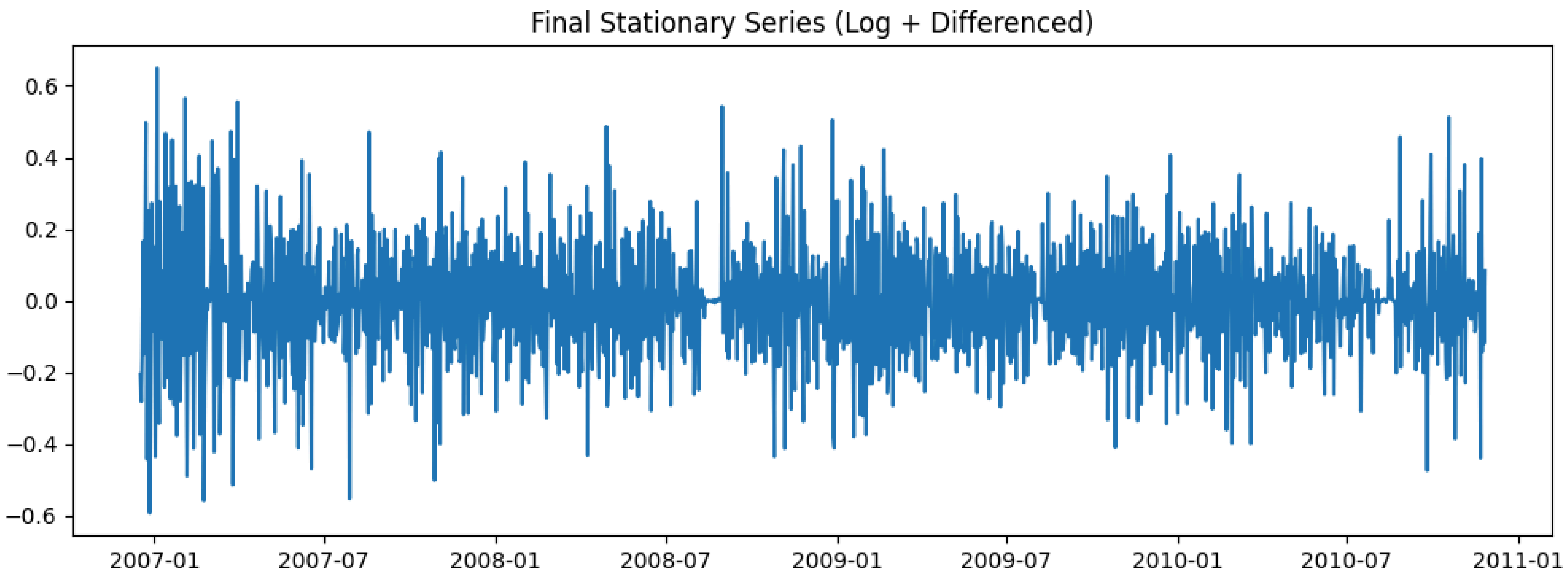
STATIONARITY

A time series is Stationary if its statistical properties do not change over time. Specifically, it must satisfy three conditions:

1. Constant Mean: The average value doesn't drift up or down over time (no trend).
2. Constant Variance: The "spread" or volatility is roughly the same throughout (no massive expansions/contractions in fluctuations).
3. Constant Autocovariance: The relationship between observations relies only on the time lag, not on when the observations were made.

Why did we use it?

- Model Requirement: Most classic forecasting models (like ARIMA) assume the data is stationary. They struggle to predict a moving target.
- Predictability: If the statistical rules of the data change tomorrow (non-stationary), we cannot use the rules from yesterday to predict it. Stationarity ensures the "rules of the game" stay the same.



This plot shows our transformed data. The fact that it oscillates around a flat zero line proves we successfully removed the trend and seasonality, making it **stationary**. However, the changing height of the spikes indicates **volatility clustering**, suggesting we should use a GARCH model to handle the changing variance.

LINEAR TIME SERIES MODELS – BASELINE MODELING

Linear models capture linear dependence between current and past observations

1. AR (Autoregressive)

- Current value depends on past observations
- Uses p lag values
  - Equation: $Y_t = c + \phi_1 Y_{t-1} + \ldots + \phi_p Y_{t-p} + \epsilon_t$

2. MA (Moving Average)

- Current value depends on past forecast errors
- Uses q lagged error terms
  - Equation: $Y_t = c + \epsilon_t + \theta_1 \epsilon_{t-1} + \ldots + \theta_q \epsilon_{t-q}$

3. ARMA (Autoregressive Moving Average)

- Combination of AR + MA
- Captures both past values and past shocks
- Used only for stationary time series

ARIMA MODEL SELECTION - FINAL LINEAR MODEL

ARIMA (Autoregressive Integrated Moving Average)

- Extends ARMA to handle non-stationary data
- Uses differencing to make the series stationary

Model Parameters

- p → AR terms (past observations)
- d → differencing order
- q → MA terms (past errors)

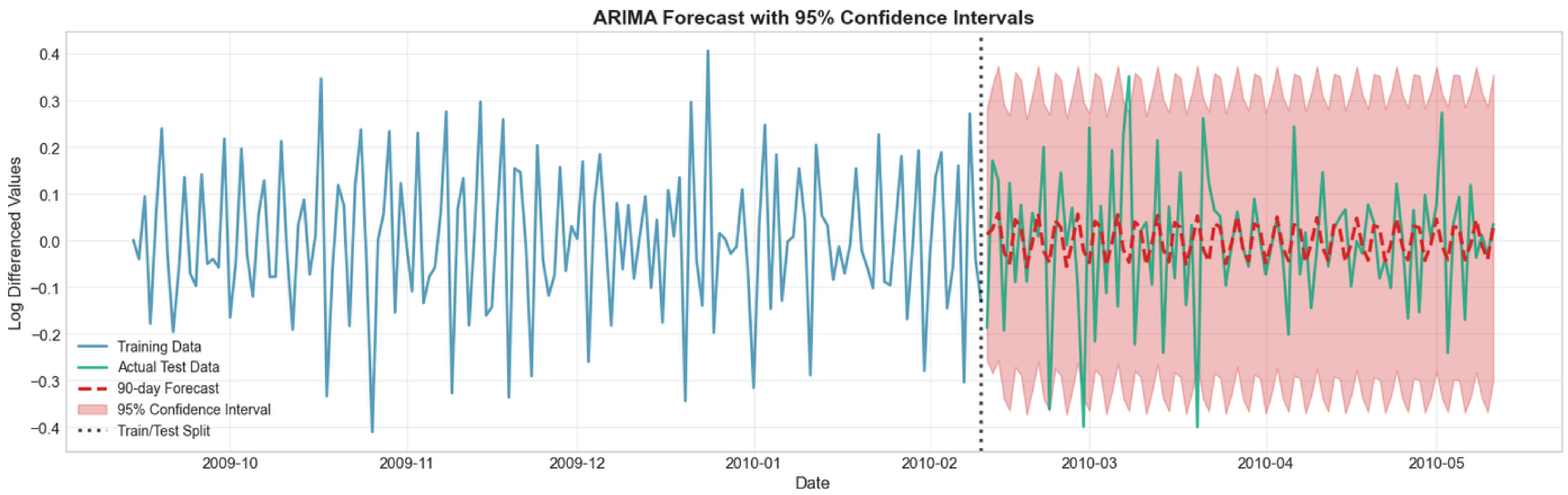
Parameter Selection

- ACF plot → determines q and PACF plot → p
- ADF & KPSS tests → check stationarity
- AIC & BIC → select best model

Final Model

- ARIMA(2,0,4)
- AIC: -1282.51
- BIC: -1242.12

ARIMA Forecast



NON-LINEAR MODEL (GARCH)

GARCH (Generalized Autoregressive Conditional Heteroskedasticity)

Purpose:

- Models time-varying volatility in electricity consumption.

Why GARCH?

- Electricity demand often shows volatility clustering
- Variance changes over time

Model Equation (GARCH(1,1))

$$\sigma t^2 = \omega+\alpha(\epsilon t-1)^\wedge 2+\beta(\sigma t-1)2$$

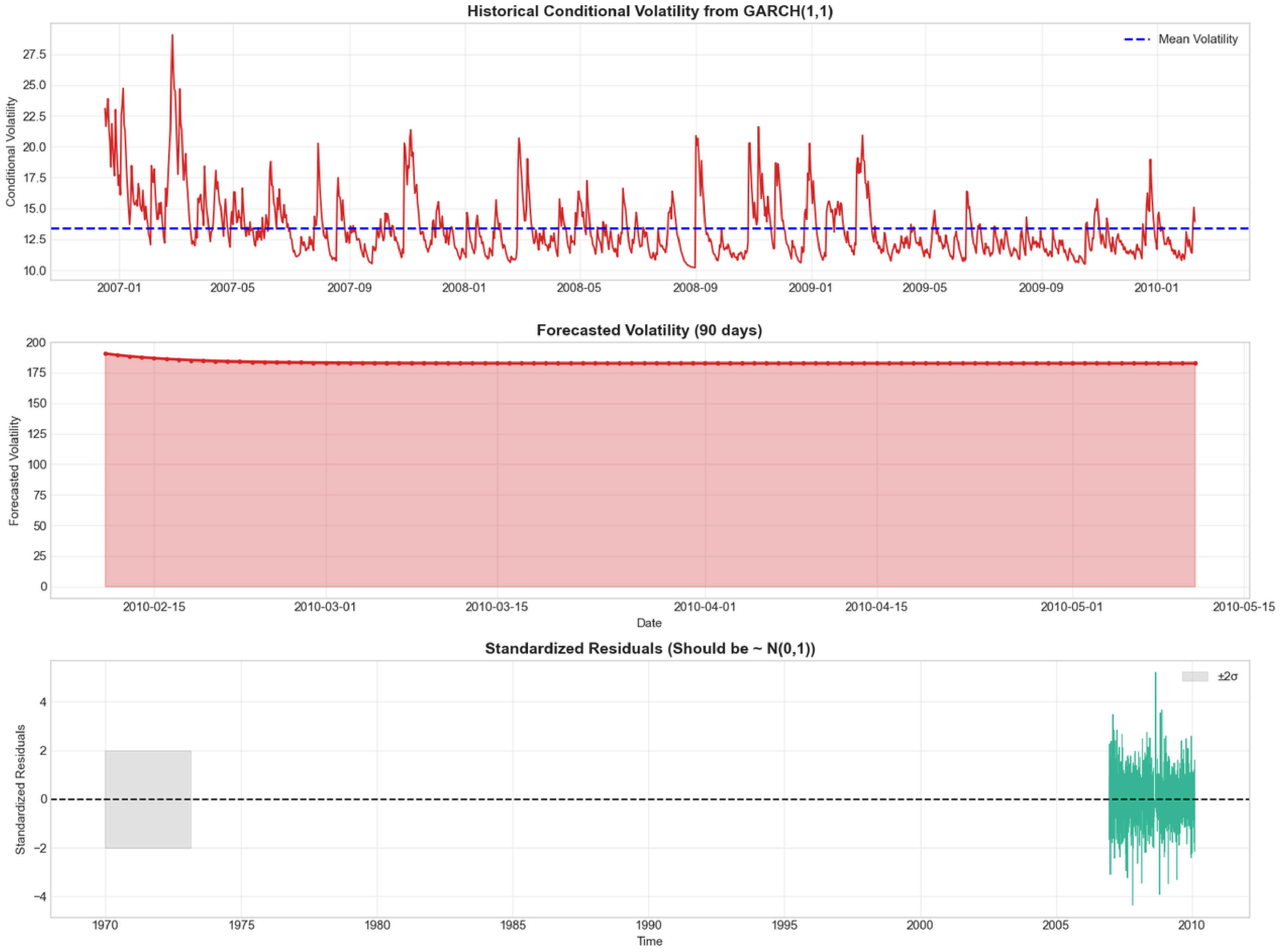
Where:

- $\sigma^2{}_t$ → current variance
- ω (omega) → constant variance term
- α (alpha) → effect of previous squared error (shock)
- β (beta) → effect of previous variance (volatility persistence)
- $\varepsilon^2{}_{t-1}$ → past squared residual

Volatility Persistence

- $\alpha + \beta = 0.848$ (< 1) → Indicates strong but stable volatility persistence, meaning volatility shocks persist for some time but gradually decay and revert to the long-run average.

GARCH RESULT ANALYSIS



- Several volatility spikes and clusters are observed, confirming the presence of volatility clustering where high-volatility periods tend to follow each other.
- The 90-day forecast shows mean reversion, with volatility gradually stabilizing toward the long-run average.
- Standardized residuals fluctuate around zero within ±2σ, indicating the model adequately captures the volatility dynamics.

Deep Learning Models

Traditional models struggle with complex patterns in time-series data.

Recurrent neural networks like LSTM and GRU are effective because they:

- Learn temporal dependencies
- Capture long-term patterns
- Handle sequential data efficiently

Input Data

- Past 168 hours (1 week) of electricity usage
- Additional engineered features:
- Hour of day, Day of week, Month, Lag values (past electricity readings), Rolling statistics (mean, min, max), Exponential moving averages

Deep Learning Model: LSTM Architecture

LSTM is a type of Recurrent Neural Network (RNN) designed to learn long-term dependencies in time-series data.
Key Idea - LSTM remembers important past information while forgetting irrelevant data.

Input to the model:

- Past 168 hours of electricity usage
- Engineered features for each hour

Model layers:

- First LSTM layer → 126 neurons
- Second LSTM layer → 64 neurons
- Final LSTM layer → 32 neurons

Additional components:

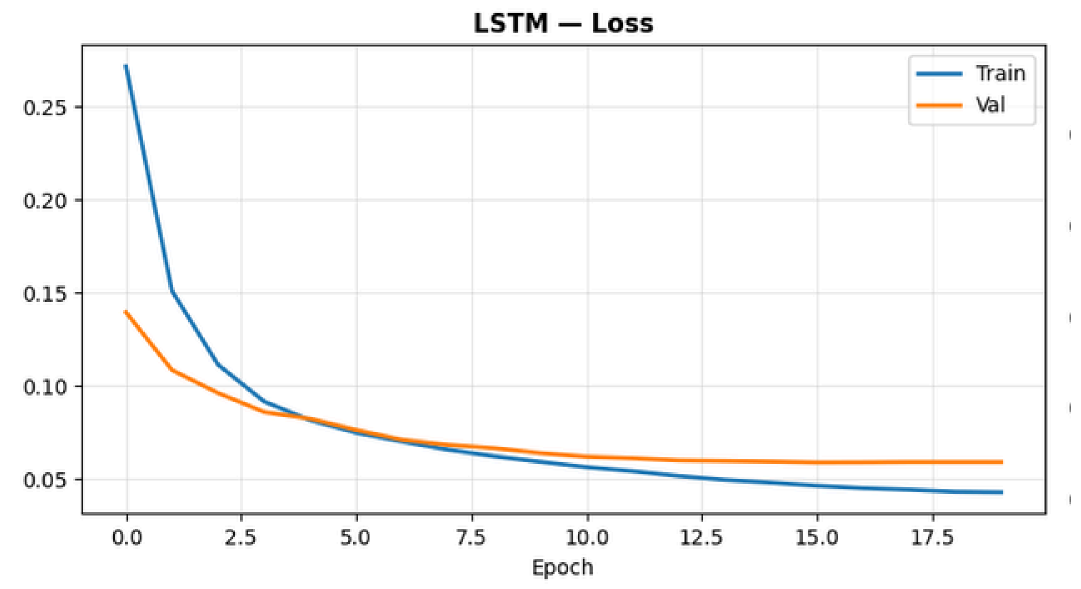
- Batch Normalization → stabilizes training
- Dropout → prevents overfitting
- Attention (SE Block) → helps the model focus on important patterns

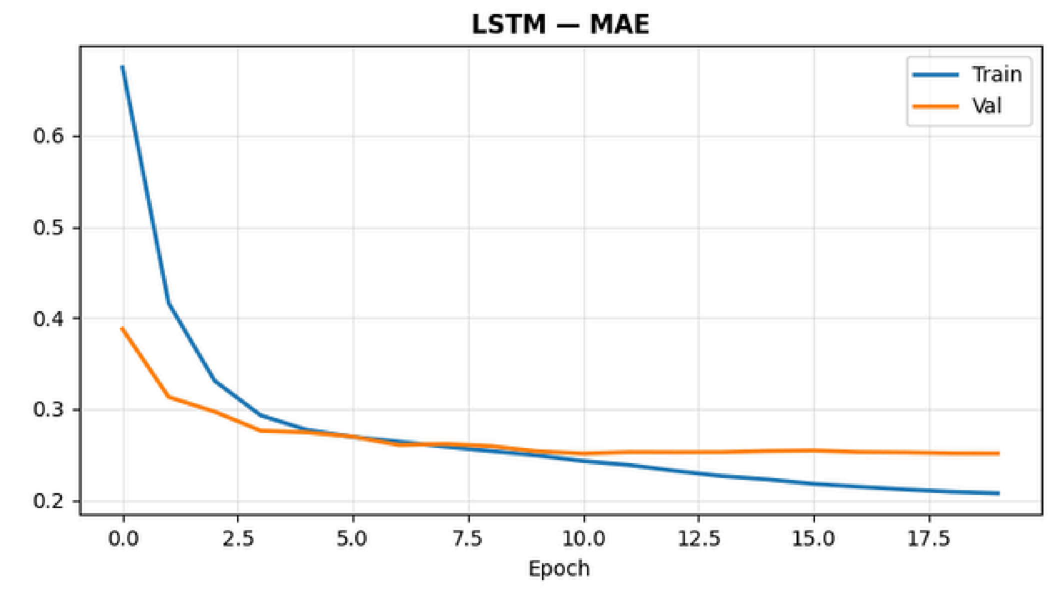
Output:

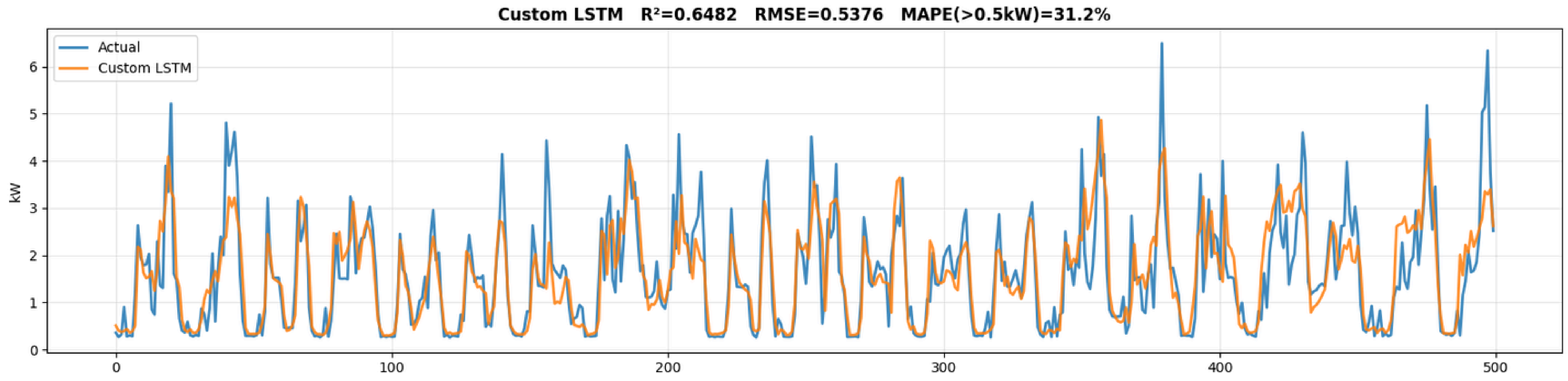
- Predicts next hour electricity consumption

Advantage of LSTM

- Excellent at learning long-term trends
- Handles complex temporal relationships







Deep Learning Model: GRU Architecture

GRU is another type of Recurrent Neural Network similar to LSTM
but simpler and faster.

GRU combines memory and update mechanisms into fewer gates.
This allows the model to:

- Train faster
- Use fewer parameters
- Maintain good prediction accuracy

Architecture used in the project
Input:

- Same 168-hour sequence data
- Same engineered features

Model layers:

- First GRU layer → 126 neurons
- Second GRU layer → 64 neurons
- Final GRU layer → 32 neurons

Other components:

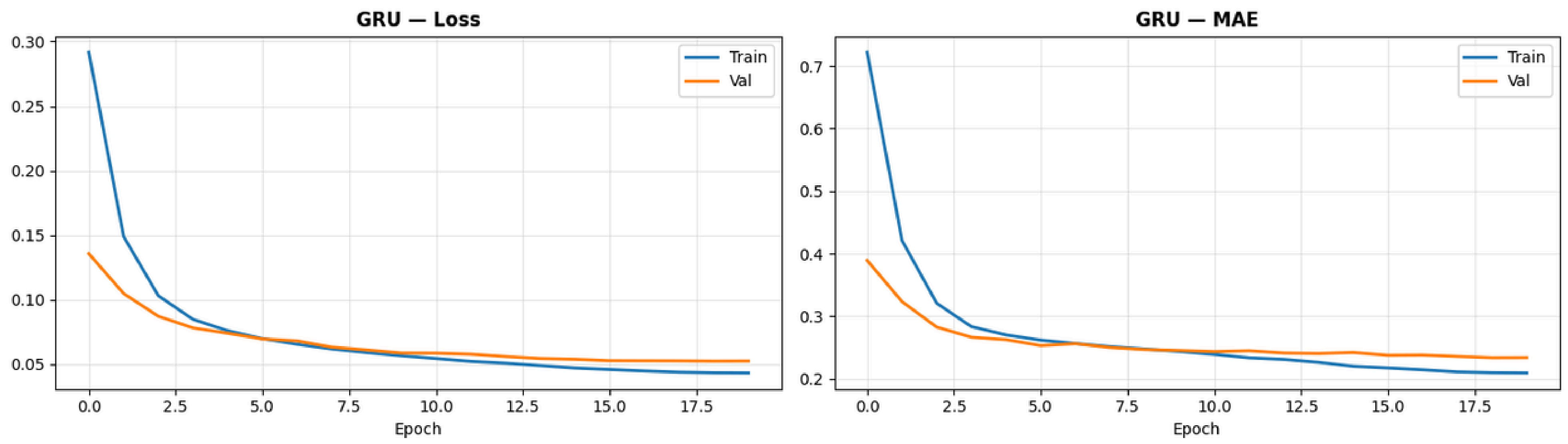
- Batch Normalization
- Dropout layers
- Attention mechanism

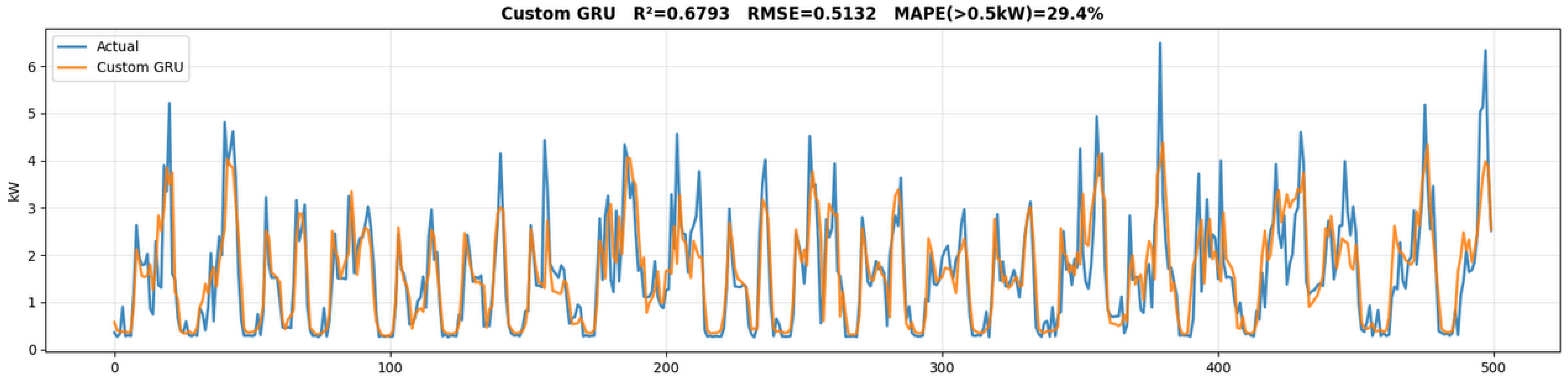
Output:

- Predicts next hour electricity usage

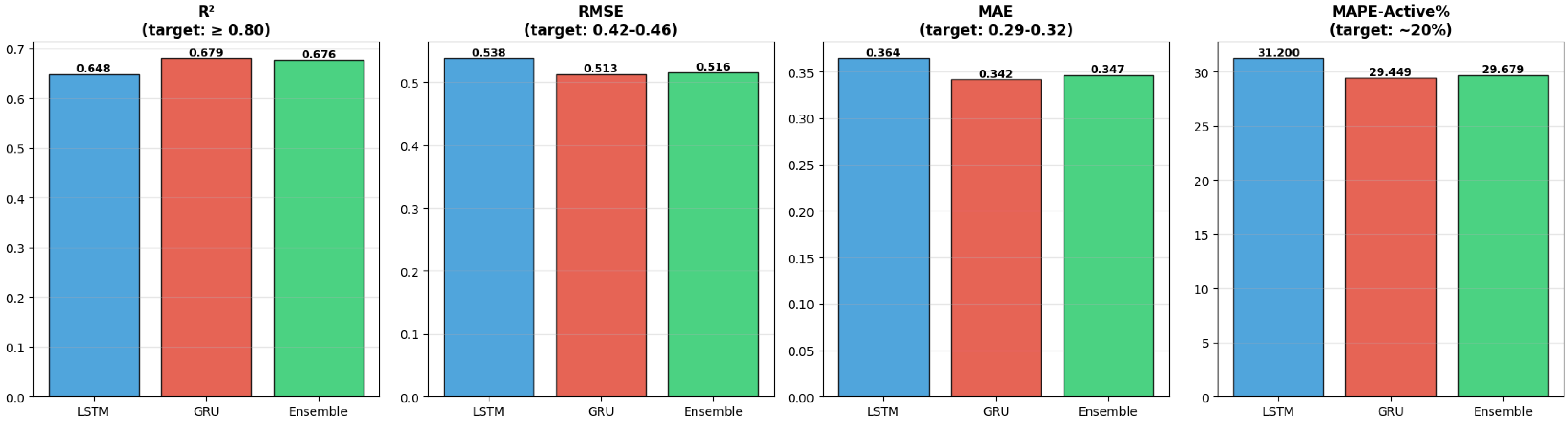
Advantage of GRU

- Faster training
- Lower computational cost
- Good performance on time-series forecasting





Model Evaluation



Models Comparison: Best Linear Model and Best Deep learning Model

Aspect	ARIMA	GRU
Model Type	Statistical	Deep Learning
Pattern Learning	Linear	Nonlinear
Training Complexity	Low	High
Feature Learning	Manual	Automatic
Temporal Pattern Handling	Limited	Strong

- ARIMA performs well for simple linear trends in electricity consumption.
- GRU captures complex temporal patterns and nonlinear relationships.
- GRU generally achieves better forecasting accuracy when data contains seasonality and complex patterns.
- Final Insight
- Traditional models like ARIMA provide interpretable baseline forecasts, while deep learning models such as GRU capture richer temporal dynamics and often produce more accurate predictions.
- Combining insights from both approaches helps in building more robust energy forecasting systems.

THANK YOU!